PlacementIQ

An adaptive multi-agent placement-readiness coach for engineering students

Abstract

Engineering students prepare for campus placements with generic material that ignores their resume, their target role and their own weak topics, and general-purpose AI chatbots give unchecked, unrepeatable advice. PlacementIQ is a multi-agent system that turns a resume and a target role into a personal, self-correcting preparation loop. A profile agent extracts skills; code computes a readiness score and the missing skills against a role-to-skill map. A planner agent writes a seven-day study plan weighted to the weakest topics, a quiz agent sets adaptive multiple-choice quizzes, and an interviewer agent runs HR, technical and project mock interviews with follow-up questions and scored feedback. A critic, built from deterministic checks followed by an LLM judge in a LangGraph generate–verify–retry loop, verifies every generated output and forces regeneration when it fails. Quiz scores update each topic's weakness, which reshapes the next week's plan. On 20 frozen evaluation cases against a single-prompt baseline, PlacementIQ produced valid plans in 100% of cases (baseline 20%) and valid quizzes in 100% (baseline 65%). It runs as one Streamlit application with one SQLite file on a free LLM tier, and fully offline in a mock mode.

Keywords: multi-agent systems, large language models, LangGraph, critic agent, adaptive learning, placement preparation, Streamlit, SQLite.

Key features

- **Resume profiling:** a PDF, DOCX or TXT resume against one of five roles gives a readiness score, matched and missing skills, and concrete resume fixes.
- **Adaptive weekly plan:** seven days of study with minutes weighted to weak topics, the reason for the plan, the shift from last week, day check-off and export to calendar (.ics) or CSV.
- **Adaptive quizzes:** topics ranked by weakness, difficulty suggested from recent scores, five questions never repeated from the student's history, and an answer review saved as notes.
- **Mock interviews:** HR, technical and project rounds; follow-ups on vague answers; each answer scored on relevance, depth, structure and clarity, with a model answer and a session summary.
- **Critic loop on every output:** free deterministic checks first, then an LLM critic; only rejected questions are rewritten, approved ones are cached, and every step is shown as a trail.
- **Closed feedback loop:** quiz results change topic weakness, quiz difficulty and the next week's plan automatically.
- **Progress dashboard:** readiness over time, study streak, 12-week activity calendar, topic mastery with trends, interview scores, and agent and critic statistics.
- **Built-in evaluation:** a resumable harness compares PlacementIQ with a single-prompt baseline on 20 frozen cases; a Results page explains each metric.
- **Practical to run:** offline mock mode, light and dark themes, phone-readable layout, free-tier Groq models, 400+ automated offline tests and CI.

Problems it solves

#	Problem	How PlacementIQ solves it
1	Students do not know how ready they are for a role or which skills are missing.	A readiness score and missing-skill list computed by code from the resume and a role-to-skill map, so it is explainable and repeatable.
2	Preparation is generic and unplanned, with no focus on personal weak areas.	A day-by-day weekly plan that gives the most time to the weakest topics and states why.
3	Study plans stay fixed while the student improves or falls behind.	Quiz scores update topic weakness and difficulty; the next plan shifts minutes accordingly.
4	AI-generated content is unreliable: invalid plans, wrong answer keys, ambiguous or duplicate questions.	Every output passes deterministic checks and a critic before it is shown; plan validity 100% vs 20%, quiz validity 100% vs 65% for a single prompt.
5	Practice repeats the same questions.	Questions are hashed and checked against everything the student was asked before; over 5 back-to-back quizzes per topic, measured for CN, DBMS and DSA, 0% of questions repeated (baseline 5.3%).
6	Little access to realistic interview practice with feedback.	On-demand mock interviews with follow-ups on vague answers (90% relevant in the evaluation), rubric scores and a model answer.
7	Progress is invisible, so motivation and course-correction suffer.	A dashboard of readiness over time, streak, activity and topic mastery.
8	Paid tools and complex infrastructure put AI coaching out of reach.	Free LLM tier, one process and one database file, and an offline mode for lab machines.